

Practice Questions: Data Frames in R Programming

Section A: Very Short Answer Questions

1. What is a data frame in R?

A data frame is a two-dimensional table in R where columns can have different data types

2. Which function is used to create a data frame in R?

`data.frame()`

3. How do you check the structure of a data frame?

`str(df)`

4. Write the command to display the first six rows of a data frame.

`head(df)`

5. How do you find the number of rows in a data frame?

`nrow(df)`

Section B: Short Answer Questions

1. Explain how a data frame is different from a matrix in R.

Data Frame: Can store different data types in different columns.

Matrix: Stores only one data type.

2. How can you add a new column to an existing data frame?

`df$Age <- c(20,21,22)`

3. Write R code to rename columns of a data frame.

`colnames(df) <- c("RollNo", "Name", "Marks")`

4. How do you extract a specific column from a data frame?

```
df$Marks
```

5. What is the use of the subset() function?

```
subset(df, Marks > 80)
```

Section C: Long Answer / Descriptive Questions

1. Explain different methods to create a data frame in R with examples.

Method 1

```
df1 <- data.frame(  
  RollNo=c(1,2,3),  
  Name=c("A","B","C"),  
  Marks=c(80,90,85)  
)
```

Method 2

```
RollNo <- c(1,2,3)  
Name <- c("A","B","C")  
Marks <- c(80,90,85)
```

```
df2 <- data.frame(RollNo, Name, Marks)
```

2. Discuss various ways to access rows and columns in a data frame.

```
df[1, ]    # First row
```

```
df[,2]     # Second column
```

```
df$Marks   # Marks column
```

```
df[1:2, ]  # First two rows
```

```
df[, c("Name","Marks")]
```

3. Explain how to handle missing values (NA) in a data frame.

```
is.na(df)
```

```
na.omit(df)
```

```
df$Marks[is.na(df$Marks)] <- mean(df$Marks, na.rm=TRUE)
```

4. Describe how to merge two data frames with examples.

```
df1 <- data.frame(ID=c(1,2), Name=c("A","B"))
```

```
df2 <- data.frame(ID=c(1,2), Marks=c(80,90))
```

```
merge(df1, df2, by="ID")
```

5. Explain sorting and ordering of data frames in R.

```
df[order(df$Marks), ]
```

```
df[order(-df$Marks), ]
```

Section D: Practical / Programming Questions

1. Create a data frame containing student details (Roll No, Name, Marks) and display it.

```
student <- data.frame(  
  RollNo=c(1,2,3),  
  Name=c("A","B","C"),  
  Marks=c(85,90,78)  
)
```

```
print(student)
```

2. Write an R program to add a new row to an existing data frame.

```
student <- rbind(student,  
  
  data.frame(RollNo=4,  
  
    Name="D",  
  
    Marks=88))  
  
print(student)
```

3. Write R code to remove a column from a data frame.

```
student$Marks <- NULL  
  
print(student)
```

4. Create a data frame and find the mean of a numeric column.

```
student <- data.frame(  
  
  RollNo=c(1,2,3),  
  
  Name=c("A","B","C"),  
  
  Marks=c(80,90,70)  
  
)  
  
mean(student$Marks)
```

5. Write a program to filter rows based on a condition.

```
student <- data.frame(  
  RollNo=c(1,2,3),  
  Name=c("A","B","C"),  
  Marks=c(80,90,70)  
)
```

```
subset(student, Marks > 75)
```